

QUESTION 1**[14 marks]**

This question is to be completed without the use of a calculator. ALL steps must be shown. Answer only will be awarded ZERO.

Simplify:

$$\begin{aligned} \text{a) } \left(\frac{2^{-2}a^5}{3a^{-1}b} \right)^{-2} &= \left(\frac{a^6}{12b} \right)^{-2} \\ &= \frac{2^4 a^{-10}}{3^{-2} a^2 b^{-2}} \quad \checkmark \\ &= \frac{16 \cdot 9 b^2}{a^{12}} \quad \frac{144 b^2}{a^{12}} \quad \checkmark \checkmark \end{aligned}$$

(3)

$$\begin{aligned} \text{b) } (3^{-1} - 2^{-1})^2 &= \left(\frac{1}{3} - \frac{1}{2} \right)^2 \\ &= \left(\frac{2-3}{6} \right)^2 \\ &= \left(\frac{-1}{6} \right)^2 \quad \checkmark \\ &= \frac{1}{36} \quad \checkmark \end{aligned}$$

(3)

$$\begin{aligned} \text{c) } \frac{27^x \cdot 9^{x+1}}{3^{5x+3}} &= \frac{3^{3x} \cdot 3^{2x+2}}{3^{5x+3}} \quad \checkmark \\ &= \frac{3^{5x+2}}{3^{5x+3}} \\ &= 3^{5x+2-5x-3} \quad \checkmark \\ &= 3^{-1} = \frac{1}{3} \quad \checkmark \end{aligned}$$

(4)

$$d) \frac{16^{\frac{3}{4}} \cdot \sqrt{3} \cdot 4^{\frac{1}{2}}}{27^{\frac{-1}{2}} \cdot 8^{\frac{2}{3}}}$$

$$= \frac{(2^4)^{\frac{3}{4}} \cdot 3^{\frac{1}{2}} \cdot 2}{(3^3)^{-\frac{1}{2}} (2^3)^{\frac{2}{3}}} \quad \checkmark \text{pr tactics}$$

$$= \frac{2^3 \cdot 3^{\frac{1}{2}} \cdot 2}{3^{-3/2} \cdot 2^2} \quad \checkmark \text{remove bracketed}$$

$$= 2^2 \cdot 3^2 \quad \checkmark$$

$$= 36 \quad \checkmark$$

(4)

QUESTION 2

[6 marks]

Solve the following equations. Show all working.

a) $3^{2x+1} - 3^{2x} = 18$

$$3^{2x} (3 - 1) = 18 \quad \checkmark$$

$$3^{2x} = 9 \quad \checkmark$$

$$3^{2x} = 3^2$$

$$2x = 2$$

$$x = 1 \quad \checkmark$$

(3)

b) $\sqrt[3]{x^2} = 25$

$$x^{2/3} = 25 \quad \checkmark$$

$$(x^{2/3})^3 = (25)^3$$

$$x^2 = 5^6$$

$\checkmark$ (working)

$$(x^2)^{1/2} = (5^6)^{1/2}$$

$$x = 5^3 = 125 \quad \checkmark$$

(3)

QUESTION 317
[16 marks]

- a) Show by completing the square that the function $2x^2 - 12x + 22$ has a minimum value of 4. Explain why you know it is a minimum value.

$$\begin{aligned} & 2(x^2 - 6x + 11) \\ &= 2[x^2 - 6x + 9 + 11 - 9] \\ &= [(x-3)^2 + 2] \\ &= 2(x-3)^2 + 4 \quad \checkmark \end{aligned}$$

$$a > 0 \quad \therefore \text{minimum} \quad \checkmark$$

- b) Solve for x by completing the square. Leave your answer in simplest surd form.

$$2x^2 + 4x - 8 = 0$$

$$2(x^2 + 2x - 4) = 0$$

$$x^2 + 2x - 4 = 0$$

$$x^2 + 2x + 1 = 4 + 1$$

$$(x+1)^2 = 5$$

$$x+1 = \pm\sqrt{5}$$

$$x = -1 \pm \sqrt{5} \quad \checkmark \checkmark$$

(4) (5)

(5)

- c) Show that there are two points of intersection between the following functions by solving for x and y :

$$y = 3x^2 - 2x - 8 \text{ and } y = 5x - 2$$

$$x^2 - 2x - 8 = 5x - 2$$

$$x^2 - 7x - 6 = 0$$

$$3x^2 - 2x - 8 = 5x - 2 \quad \checkmark$$

$$3x^2 - 7x - 6 = 0$$

$$(3x + 2)(x - 3) = 0$$

$$x = -2/3 \quad \checkmark \quad \text{or} \quad x = 3 \quad \checkmark$$

$$\therefore y = 5(-2/3) - 2$$

$$= -16/3 \quad \checkmark$$

$$y = 13 \quad \checkmark$$

(7)

QUESTION 4

[18 marks]

- a) The general term of $5 \quad 12 \quad 29 \quad 48 \quad 77 \dots$ is $T_n = 3n^2 + 2$

Show by using a minimum of TWO calculations whether you agree with the statement or not. Explain your conclusion.

$$T_1 = 3(1)^2 + 2$$

$$= 3 + 2$$

$$= 5 \quad \checkmark$$

$$T_1 = 3(2)^2 + 2$$

$$= 3(4) + 2$$

$$= 14 \quad \checkmark$$

Not true: Term 2 is 12 not 14

(3)

b) Consider the sequence:

$$\frac{2}{3} \quad \frac{1}{3} \quad \frac{1}{6}$$

- (i) Extend the sequence by 3 terms
- (ii) Determine the general term
- (iii) Determine T_{10}

$$(i) \quad \frac{1}{12} \quad \frac{1}{24} \quad \frac{1}{48} \quad \checkmark$$

$$(ii) \quad T_n = \left(\frac{2}{3}\right) \left(\frac{1}{2}\right)^{n-1} \quad \checkmark \quad \checkmark$$

$$(iii) \quad T_{10} = \left(\frac{2}{3}\right) \left(\frac{1}{2}\right)^9 \quad \checkmark \\ = \frac{1}{168} \quad \checkmark$$

~~15)~~

c) Consider the quadratic pattern:

-9 -6 1 x 27

- (i) Give the value of the second difference
- (ii) Find the value of x
- (iii) Determine which term in the sequence has a value of 397

(i) 4 ✓

(ii) $x - 8 = 4$ ✓
 $x = 12$ ✓

(iii) $2a = 4$ $3(2) + b = 3$ $2 - 3 + c = -9$
 $a = 2$ ✓ $b = -3$ ✓ $c = -8$ ✓

$$T_n = 2n^2 - 3n - 8$$

$$397 = 2n^2 - 3n - 8$$

$$0 = 2n^2 - 3n - 405$$

$$n = -27\frac{1}{2} \quad n = 15$$

$$\therefore 397 \text{ is } T_{15}$$

(10)